

Image Generation with Conditional GAN (cGAN)

Abstract:

Generative Adversarial Networks (GANs) have emerged as powerful models for synthetic data generation. This project implements a Conditional GAN (cGAN) to generate class-specific images using the Fashion-MNIST dataset. The model incorporates class embeddings into both the Generator and Discriminator, enabling controlled image generation. Several stabilization techniques such as label smoothing, spectral normalization, and balanced training updates are employed to improve performance. The model achieves a final Frechet Inception Distance (FID) score of 10.91, indicating good similarity between generated and real images. Additionally, latent space exploration through interpolation and class morphing demonstrates meaningful feature learning and smooth transitions. Overall, the model successfully generates realistic and diverse fashion images while maintaining stable training behavior.

Introduction:

Generative Adversarial Networks (GANs), introduced by Goodfellow et al., are widely used for generating realistic synthetic data. A GAN consists of two competing neural networks: a Generator that produces fake samples and a Discriminator that evaluates their authenticity. Through adversarial training, both networks improve iteratively.

This project focuses on implementing a Conditional GAN (cGAN), where class label information is incorporated into both networks. This allows the Generator to produce images belonging to a specific class, making the model more controllable and interpretable. The Fashion-MNIST dataset is used due to its complexity compared to MNIST, providing a more challenging benchmark.

Dataset:

The Fashion-MNIST dataset consists of grayscale images representing different clothing categories. It contains:

- 60,000 training images
- 10 classes (e.g., T-shirt, Trouser, Sneaker, Bag)
- Image size: 28×28 pixels

Before training, images are normalized to the range $[-1, 1]$ to match the Generator's Tanh activation function. The dataset provides sufficient diversity for evaluating generative models.

Methodology:

1. Generator Architecture

The Generator takes two inputs:

- A random noise vector of dimension 100
- A class embedding vector of dimension 10

These inputs are concatenated and passed through multiple transposed convolution layers to progressively upscale the feature maps from 1×1 to 28×28 . Batch normalization and ReLU activations are used in intermediate layers, while Tanh activation is applied at the output layer.

2. Discriminator Architecture

The Discriminator receives:

- An input image
- A class embedding reshaped to match image dimensions

These are concatenated and passed through convolutional layers with LeakyReLU activation. Spectral normalization is applied to improve stability. The output is a single scalar indicating whether the image is real or fake.

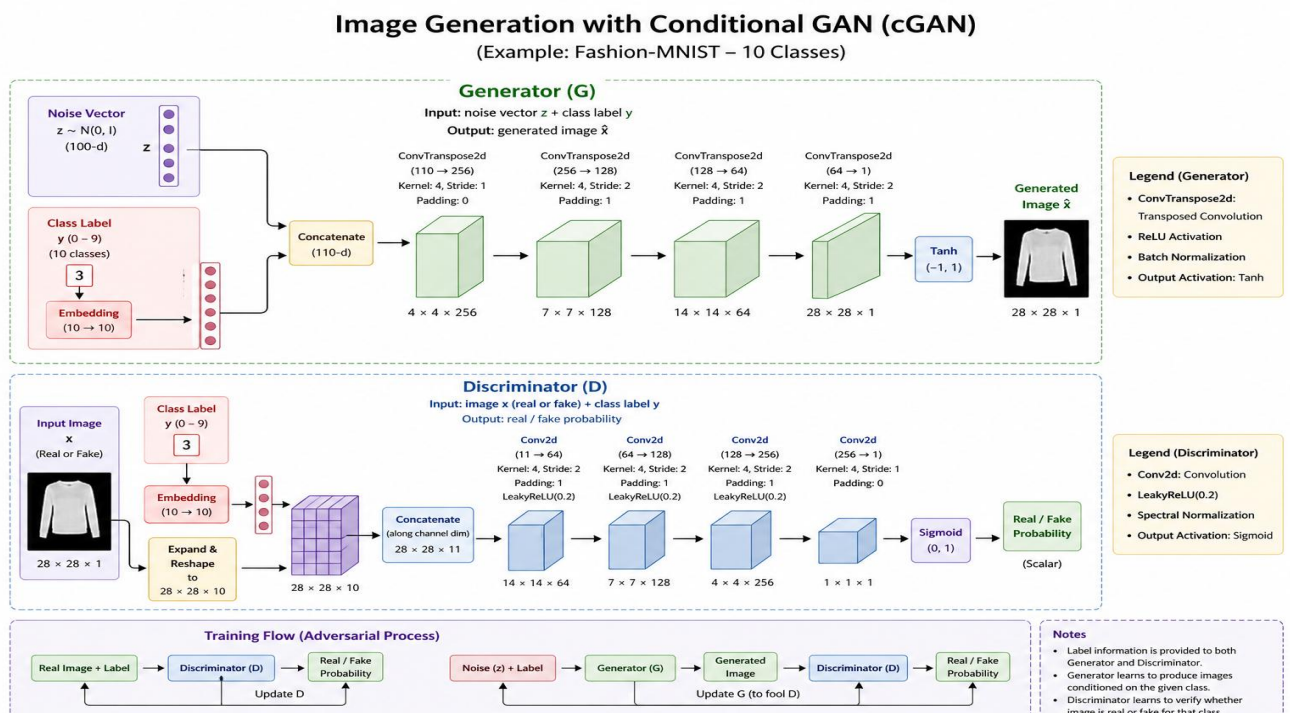


Figure 1: Conditional GAN architecture

3. Training Strategy

To stabilize training, the following techniques are applied:

- **Label Smoothing:** Real labels sampled between 0.9 and 1.0
- **Balanced Updates:** Discriminator updates skipped if loss < 0.3
- **Optimizer:** Adam ($lr = 2e-4$, $\beta_1 = 0.5$, $\beta_2 = 0.999$)
- **Fixed Noise:** Used to track visual progress

The model is trained for 100 epochs.

Results:

1. Training Loss

The Generator and Discriminator losses show stable convergence without significant oscillations.



Figure 2: Training loss curves

2. FID Evaluation

The Frechet Inception Distance (FID) is used to evaluate model performance.

- Epoch 1 → **12.30**
- Epoch 50 → **11.81**
- Epoch 100 → **10.91**

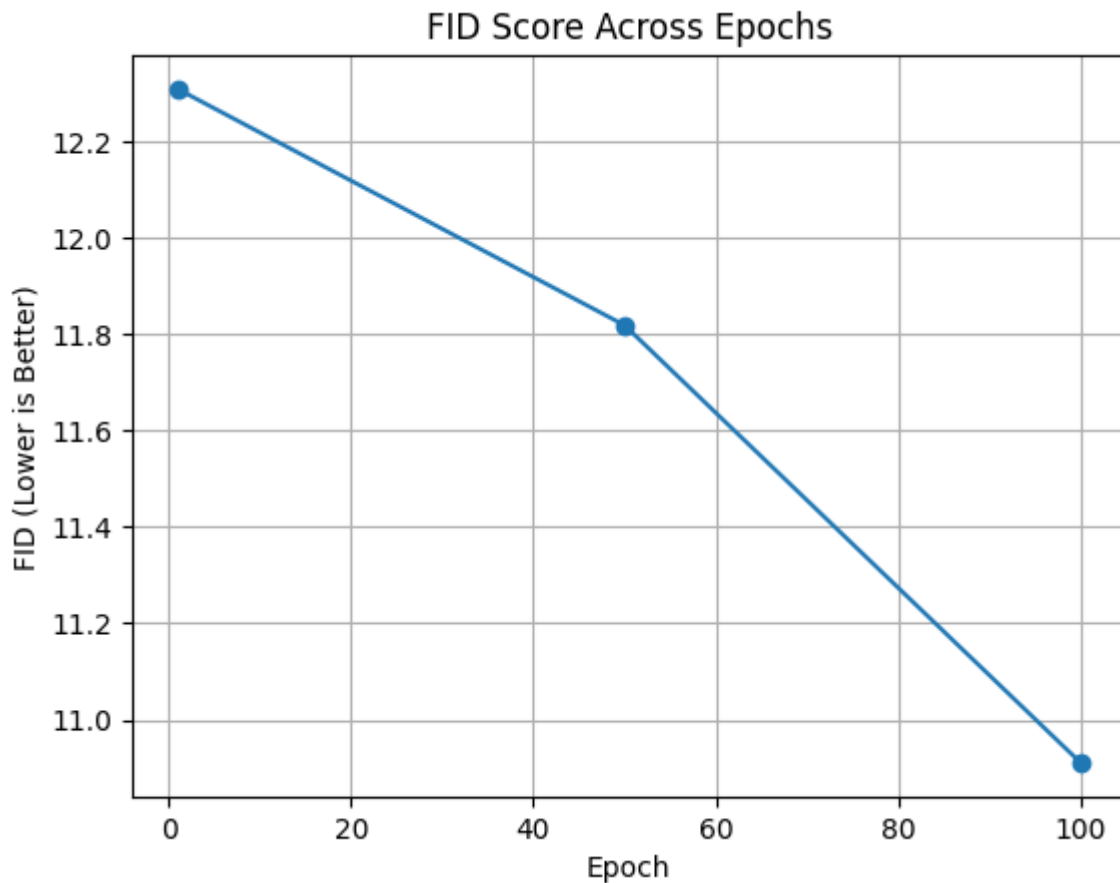


Figure 2: FID score across epochs.

Analysis:

The decreasing FID score indicates continuous improvement in the Generator's ability to produce realistic images. The final FID score of 10.91 falls within the acceptable range for GAN performance on Fashion-MNIST.

The training losses remain stable throughout the process, indicating balanced adversarial learning. No major instability or divergence is observed.

Mode collapse is not evident, as the model generates diverse outputs across different classes. This is further supported by latent space exploration.

1. Latent Space Interpretation

- **SLERP Interpolation**

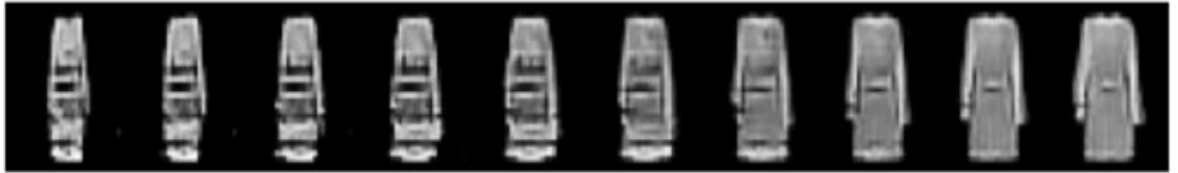


Figure 3: Smooth transition between latent vectors.

- **Class Morphing**



Figure 4: Transition between different classes.

These results demonstrate that the model has learned a structured and continuous latent space.

Conclusion:

This project successfully implements a Conditional GAN for Fashion-MNIST image generation. The model demonstrates stable training, effective class conditioning, and good image quality. The decreasing FID score confirms improved performance over time. Latent space analysis further validates the model's ability to learn meaningful representations. Overall, the approach achieves the objectives of generating realistic and diverse images.

References:

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5. Goodfellow et al., "Generative Adversarial Networks", 2014